

Neural State Architecture: Decoupling Semantic Representation from Information-Flow Policy Algebra in Large Language Models

Neural State Architecture Research Group

Adam Lapworth

Current Large Language Models (LLMs) rely on fragile external text wrappers to enforce security and privacy policies. Because continuous activations flow through unrestricted continuous spaces, external wrappers are vulnerable to prompt injections and activation probes. We introduce **Neural State Architecture (NSA)**, a computational paradigm that embeds typed activations $h = (m, \sigma_h, \sigma_s, v)$ and state transition operators (w, V) directly into neural forward passes. NSA enforces information security policies via a mathematical product state lattice algebra, making unauthorized information flow mathematically impossible at the attention layer via rigorous Hard Policy Semantics, while introducing Risk-Weighted Soft Semantics for post-hoc retrofitting. We validate NSA across synthetic benchmarks, demonstrating 0.00 % prompt injection hijack rate when structural constraints are combined with value alignment.

Introduction

Modern Large Language Models (LLMs) process all contextual inputs within a single untyped vector space R^d . This architecture fails to provide mathematical guarantees regarding information boundaries.

Definition 1 (Typed Product Lattice). *Let Σ be a product state space defined over multiple bounded join-semilattices (e.g., Security, Confidence, Provenance, License). Each token hidden state at layer l is represented as a coupled quad-tuple: $h_i^{(l)} = (m_i^{(l)}, \sigma_{h,i}^{(l)}, \sigma_{s,i}^{(l)}, v_i^{(l)})$ where m_i represents the semantic activation, σ_h represents the hard structural state (Confidentiality, Integrity), σ_s represents the soft operational state (Confidence), and v represents the value/preference layer.*

Information-Flow Governance & Typed Declassification

State-aware attention calculates query-key compatibility by composing semantic similarity with state transition permission:

$$A_{ij} = \text{Softmax} \left(\frac{Q_i K_j^T}{\sqrt{d_k}} + M_{\text{state}}(\sigma_i, \sigma_j) \right)$$

When token j possesses a security clearance higher than token i , $M_{\text{state}}(\sigma_i, \sigma_j) \rightarrow -\infty$, reducing $A_{ij} \rightarrow 0$ (Hard Policy Semantics). For retrofitted unconstrained LLMs, applying a

rigid mask causes severe out-of-distribution activation spikes, necessitating a Risk-Weighted Soft Policy mask.

Typed Declassification Primitive

Downward reclassification algebraically requires passing an explicit typed capability:

$$D:(\sigma, c_D) \rightarrow \sigma'$$

where $\text{Valid}(c_D, \sigma, \sigma')=1$ and $c_D=(\text{issuer}, \text{purpose}, \text{scope}, \text{expiry}, \text{max downgrade})$. This transforms declassification into a formal, auditable computational primitive rather than an unconstrained vulnerability.

Formal Non-Interference Proof

Theorem 1 (Non-Interference Bounds). *Let m_{src} be a secret semantic token activation with state $\sigma_{\text{src}}=\text{CONFIDENTIAL}$, and let m_{dst} be a public output activation with state $\sigma_{\text{dst}}=\text{PUBLIC}$. Under hard state masking ($M_{\text{state}}=-\infty$), the mutual information satisfies:*

$$I(m_{\text{src}}; m_{\text{dst}} \mid \sigma)=0$$

Proof. By definition of state-aware attention, when $s_{\text{src}} \not\leq s_{\text{dst}}$, the attention weight $A_{\text{dst}, \text{src}}=0$. Consequently, the value aggregation step yields:

$$v_{\text{out}} = \sum_{j \neq \text{src}} A_{\text{dst}, j} V(m_j)$$

Because v_{out} is functionally independent of m_{src} , the conditional probability $P(m_{\text{dst}} \mid m_{\text{src}}, \sigma)=P(m_{\text{dst}} \mid \sigma)$. Thus:

$$I(m_{\text{src}}; m_{\text{dst}} \mid \sigma)=H(m_{\text{dst}} \mid \sigma)-H(m_{\text{dst}} \mid m_{\text{src}}, \sigma)=0$$

□

Empirical Evaluation

We validate the theoretical guarantees of NSA via a synthetic injection-attack benchmark evaluating the random-guess base rate (2%):

Conclusion

Neural State Architecture demonstrates that security and privacy policies do not require fragile external wrappers. By embedding typed product state lattices directly into continuous Transformer activations alongside robust capabilities and transition operators, NSA achieves provable information isolation.